

Question 1

Not yet answered

Marked out of 14.00

Q 10 for quiz

*Note that you can upload your python notebook in the upload box below.

1. Coordinate frame 1 is rotated by 30 degrees clockwise around the y axis and displaced by (0,1,1) with respect to coordinate frame 0.

Provide answers correct to 3 decimal places.

1. Show the homogenous transformation matrix

$$T_1^0$$

1. Point

$$P^1$$

= [1,2,1]. What is

$$P^0$$

?

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1. Coordinate frame 2 is rotated by 60 degrees anticlockwise around the z axis and displaced by (2,1,0) with respect to axis 1.

1. Calculate

$$T_2^1$$

2. Calculate

$$T_2^0$$

Question 2

Not yet answered

Not graded

You can upload your python notebook here if you have one. Upload an empty file here if you want the question to be marked as complete but don't have anything to upload.

Maximum file size: 2 GB, maximum number of files: 1





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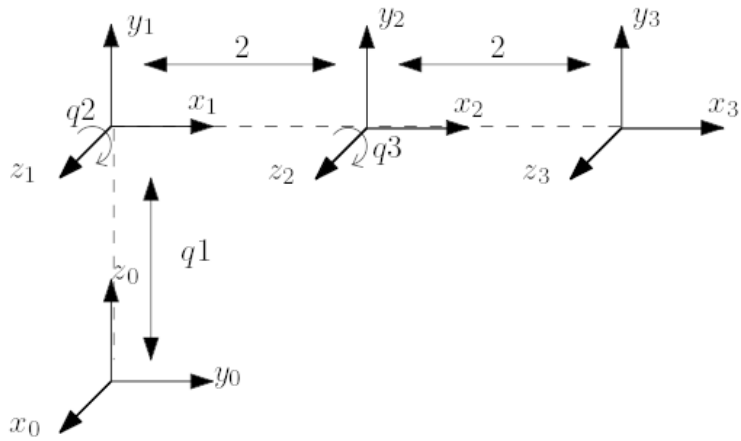
You can drag and drop files here to add them.

Question 3

Not yet answered

Marked out of 6.00

The PRR arm used for this question is shown below



Fill in the corresponding table of DH parameters (using degrees, not radians)

i	a_i	α_i	d_i	θ_i
1				
2				
3				

Question 4

Not yet answered

Marked out of 13.00

You have an RPP arm with the following transformation matrices.

Work out

$$J_{\omega}$$

and

$$J_v$$

$$T_1^0 = \begin{pmatrix} \cos(q1) & -\sin(q1) & 0 & 0 \\ \sin(q1) & \cos(q1) & 0 & 0 \\ 0 & 0 & 1 & d1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_2^0 = \begin{pmatrix} \cos(q1) & 0 & -\sin(q1) & 0 \\ \sin(q1) & 0 & \cos(q1) & 0 \\ 0 & -1 & 0 & d1 + q2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$T_3^0 = \begin{pmatrix} \cos(q1) & 0 & -\sin(q1) & -\sin(q1)q3 \\ \sin(q1) & 0 & \cos(q1) & \cos(q1)q3 \\ 0 & -1 & 0 & d1 + q2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Note that you should format terms as $x\cos(y)$ where appropriate - like $q1\cos(q3)$ for example.

$$J_{\omega}$$

=

$$J_v$$

=

What is the linear velocity of the endpoint if joint 1 is rotated to

$$\pi/5$$

radians and is rotating at 2 radians per second, joint 2 is extended to 2 m and is extending at 1 m/s, and joint 3 is extended to 1 m and is extending at 2 m/s. d1 is a fixed value of 3 m as it does not represent a joint variable.

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Question 5

Not yet answered

Not graded

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Question 6

Not yet answered

Marked out of 5.00

We talk of a robot as an embodiment of a perception-action loop.

For each of the following, is it best described as an example of sensing, planning, or acting?

Predicting future stock prices

Vocalising an instruction to a collaborating human

Determining which crops are ready to be harvested

Identifying regions of congested traffic on a road network

Feeling if the ground ahead of the robot is stable

Determining the best way to divide up a search area between a team of mobile robots

Sending a signal to another robot with information of what it is planning on doing

Choosing the best move to make in a game of chess

Identifying the dirtiest spots on a floor to be cleaned

Moving the arms of a humanoid to ensure balance

Question 7

Not yet answered

Marked out of 5.00

Consider the following second-order system that we want to control:

$$2\ddot{x}_1 + 4\dot{x}_1 - u = 0$$

with

$$y = \dot{x}_1$$

By introducing a change of variables, rewrite this as a first-order system:

$$\dot{x} = Ax + Bu$$

, where

$$\dot{x} = (\dot{x}_1, \dot{x}_2)^T$$

$$A =$$

$$B^T =$$

Which of the following statements are true?

The system is linear: ☐

The system is time-variant: ☐

The system is controllable: ☐ because the rank of the controllability matrix is ☐

The system is observable: ☐ because the rank of the observability matrix is ☐

Question 8

Not yet answered

Marked out of 6.00

Consider the second-order system that we want to control:

$$\dot{x}_1 = 4x_2$$

$$\dot{x}_2 = 2u$$

With

$$x_1(0) = 1$$

,

$$x_2(0) = 0$$

,

where

$$x_1$$

is the position of the system and

$$u$$

is the control variable, and the cost function is given by

$$J(u) = A(x_1(3) - 4)^2 + B(x_2(3))^2 + \frac{C}{2} \int_0^3 [u(t)]^2 dt$$

By introducing an appropriate Hamiltonian, find the control and state trajectory that minimises the cost function for

$$A = 3, B = 1, C = 1$$

.

From this, we have that

$$x_1(t) =$$

$$t^3 +$$

$$t^2 +$$

$$t +$$

Now assume we set

$$A = 1, B = 0, C = 5$$

. Which of the following statements are true?

We care more about the velocity than the position of the system:

We would have an arbitrary final velocity:

The final position should be close to 1:

We care more about the integral cost than the terminal cost:

Question 9

Not yet answered

Marked out of 6.00

Consider the following matrix game, where the given payoffs are to the column player, who is the maximising player. The row player is minimising, and receives the negative of the shown reward:

$$\begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & 3 \\ 2 & 2 & 1 \end{pmatrix}$$

The Nash equilibrium policy for the column player, where each number is the probability of picking that column, is given by

 , ,

The payoff of the game is .

Which of the following statements are true?

This game is zero-sum:

The game is shown here in extensive form:

The game shown here is a discrete game because the payoffs are all integers:

The Nash equilibrium is the highest payoff that any player can receive in the game:

Question 10

Not yet answered

Marked out of 5.00

A robotic spacecraft needs to measure the distance

$$x$$

to a space station that it must dock with. The robot takes multiple readings to try and determine the distance before initialising the docking procedure. Assume neither the spacecraft nor the station moves after each reading, and so

$$x_t = x_{t-1}$$

. The noisy readings of the distance are given by

$$z_t = x_t + \delta_t$$

, where the noise is described by the process

$$\delta_t \sim N(0; 0.2)$$

Assume the current estimate is

$$x_0 = 0.79m$$

, and

$$P_0 = 1$$

. Now the spacecraft receives a reading of

$$z_1 = 0.92m$$

. Using a Kalman filter, obtain the updated mean and variance of the spacecraft's estimate to the station.

After the update

$$x_1 =$$

and

$$P_1 =$$

.

The Kalman gain

$$K =$$

.

Which of the following statements are true about Kalman filters?

The value of

$$x$$

in the predictor step

$$x_t^-$$

is always smaller than the value of

$$x$$

in the corrector step

x_t

:

A Kalman filter cannot be directly applied to a problem if the system [dynamics](#) are not known:

A Bayes filter is a Kalman filter where the distribution over

x

does not have to be Gaussian:

A particle filter is a Bayes filter where we use particles to approximate the mean and variance of the distribution: